Joint Venture Formation in Procurement Auctions

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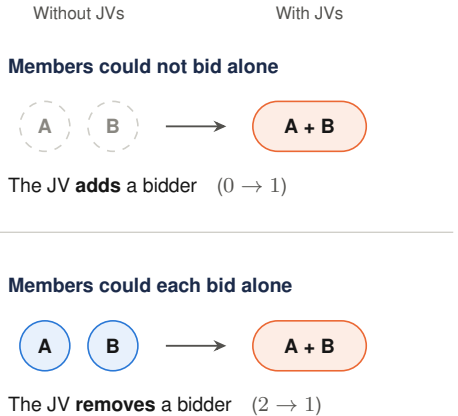
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Joint bidding: more competition, or less?

- Firms may **bid jointly** in many auctions
- **For:** projects too large for one firm
- **For:** synergies from complementary skills
- **Against:** one bid instead of two
- Practical test: *could the members bid alone?*



What would happen without joint bidding?

We cannot observe the same auction **both with and without JVs**.

Predicting the outcome without JVs requires three elements:

1. Members' participation

Would they bid alone or stay out?

2. Scores and costs

What scores and costs would members have bidding alone?

3. Rivals' responses

Who else would enter, and how would bids change?

We need a model linking JV formation and bidding.

This paper

Data	Japanese public works auctions, 2006–2022, with **JV members identified**
Model	Formation of JVs, then a scoring auction with a price floor
Key result	Equal sharing $\Rightarrow$ a **unique stable partition**, found by a greedy algorithm
Findings	1. Mean effective cost is no lower than the members' solo average More dispersed draws can make JVs very competitive 2. Firms that have partnered before are more likely to partner again 3. Most firms lack a partner with whom both would gain from forming a JV 4. In larger projects, a ban on JVs raises prices and lowers technical scores 5. With a binding price floor, prices alone miss the effect

▸ Related literature

Roadmap

1. Setting and data
2. Model
3. Estimation and results
4. Counterfactual: banning joint ventures

How the procurement auction works

1. **Tender notice:** eligibility, JV permission and scoring rules are announced
2. **Qualification:** firms apply alone or register a JV by the deadline
   A JV is assessed as one bidder and submits one proposal and one price
3. **Technical proposals:** eligible bidders submit proposals, which are scored s
4. **Sealed prices:** bidders then submit b, without observing rivals' prices
   The reserve R (engineer's estimate) and realized bidder count are not observed
5. **Award:** the highest s/b among acceptable bids wins and is paid b
   All bidders' prices and scores are published after the award

Price limits: bids above R are invalid; bids below τR trigger an investigation.

The investigation threshold acts as a price floor in practice.

▸ Size thresholds for JVs

From the full panel to 396 formation auctions

Full panel: 86,485 auctions in seven regional bureaus, 2006–2022.

Auctions	Restriction
547	Competitive auctions at or above bureau–work-category size cutoffs
477	All listed solo bidders and JV members identified
462	Every observed JV has two members
396	Every listed participant belongs to the potential-entrant set

- **Size cutoff**: keep projects at least as large as the smallest observed JV project in each region and work category
- **Potential entrants**: firms active in the same bureau and work category in the previous **3 years**, on a project **at least one-tenth as large**

▸ Construction details

Prices, scores, and auction outcomes

Seven regional bureaus, 2006–2022: published prices, technical scores, and winners.

Full bid panel

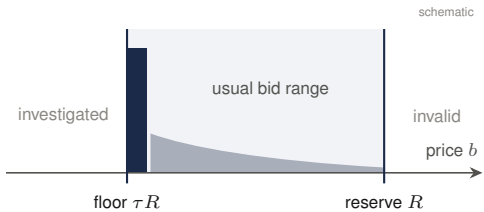
Auctions	86,485
Bidder records	574,553

Sample of 396 large auctions

- 6,159 solo listings and 312 JV listings
- **1.8%** of valid bids fall below the floor
- Win rates: solo **5.9%**, JV **10.0%**
- JV win-rate gap: **+4.1 pp** (s.e. 2.1)

JV versus solo bidders within the same auction.

Prices in the 396 large auctions



56% of valid bids **75%** of winning bids

are at or below 101% of the floor

Stylized facts

Formation pattern	Evidence
Repeated partnerships	167 of 312 JVs are between past partners
Prior contact	296 of 312 JVs have prior contact; 19.6% of potential pairs do
Concentration	JVs appear in 127 of the 396 auctions

Model assumptions

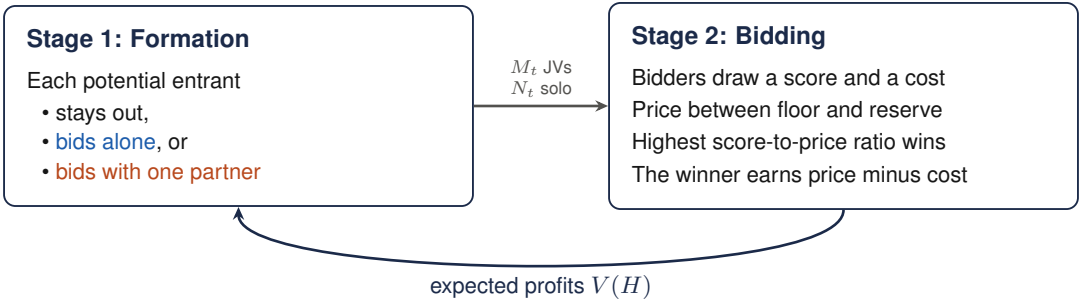
- **Repeated partnerships**: past joint bids lower additional JV costs.
- **Prior contact**: pairs with contact are always available.
  Other pairs are independently available with the same estimated probability.
- **Concentration**: an auction-level shock shifts all pairs' JV payoffs.
  Included in estimation, omitted from the simplified model presentation.

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The model in one picture

One auction t: engineer's estimate R_t, price floor $\tau_t R_t$, potential entrants $\mathcal{N}_t$.



H: beliefs before formation about the numbers of **rival JVs and solo bidders**.

The estimated model also includes auction-specific unobserved shocks in the formation stage.

▸ Timing and information

Bidding: firms differ in cost and score strength

A solo bidder i in auction t:

$$\log s_{it} = \mu_t + \sigma_i + e_{it} \qquad \text{score}$$
$$\log(c_{it}/R_t) = \delta_t + q(\log s_{it}) - \alpha_i + u_{it} \qquad \text{cost}$$

- (α_i, σ_i): persistent **bidding strength**
- (e_{it}, u_{it}): the bidder's **type**, drawn after formation
- (μ_t, δ_t): auction conditions, known to firms
- $q(\cdot)$: a higher score costs more

Strength of a JV:

$$a_{ij} = \big(\max\{\alpha_i, \alpha_j\},\ \max\{\sigma_i, \sigma_j\}\big)$$



plus JV-specific types and $q(\cdot)$

Output: expected profits $\pi_i^S(m,n)$ and $\pi_{ij}^J(m,n)$ for every firm and every pair.

Formation: three options

Stay out	0
Bid alone	$u_i^S = \dfrac{V_i^S}{\sigma} - f - \lambda_S g_i - \beta_S \ell + \varepsilon_i$
Bid with j	$u_{ij}^J = \dfrac{1}{2}\left[\dfrac{V_{ij}^J}{\sigma} - K\, e^{-\gamma h_{ij}} - \lambda_J g_{ij}\right] - f - \beta_J \ell + \nu_{ij}$

Capacity: K_i is the largest past solo project; a JV pools $K_i + K_j$.

Shortfalls: $g_i = \max\{0, \log(R/K_i)\}$, $\quad g_{ij} = \max\{0, \log(R/(K_i + K_j))\}$.

Project size: $\ell = \max\{0, \log(R/(3 \text{ bn yen}))\}$; direct effects beyond capacity.

- $V_i^S = \sum_{m,n} H^S(m,n)\, \pi_i^S(m,n)$: expected profit, given **beliefs** H about rivals
- f: participation cost. $\quad K\, e^{-\gamma h_{ij}}$: cost of a JV, with $\boldsymbol{h_{ij} = \log(1 + \text{past joint bids})}$
- ε_i, ν_{ij}: firm and pair shocks. $\quad \sigma$: s.d. of ε in yen

Equal sharing: the two members receive the same payoff from their JV.

Equal sharing gives a unique stable partition

Partition: who stays out, who bids alone, and who bids with whom.

Stable partition: no firm, and no pair of firms, wants to deviate.

Proposition 1. Given beliefs, there is exactly one stable partition (with probability one).
A **greedy algorithm** finds it: form the best acceptable pair, remove both firms, repeat.

Why

Globally ranked pairs (Abraham et al. 2008)

$\mathcal{E}$: available pairs, $\mathcal{E}_i = \{e \in \mathcal{E} : i \in e\}$.

$$\exists\, w : \mathcal{E} \to \mathbb{R} \quad \text{such that}$$
$$e \succeq_i f \iff w(e) \geq w(f), \quad \forall i, \quad \forall e, f \in \mathcal{E}_i.$$

In our model, equal net payoffs give such a w:

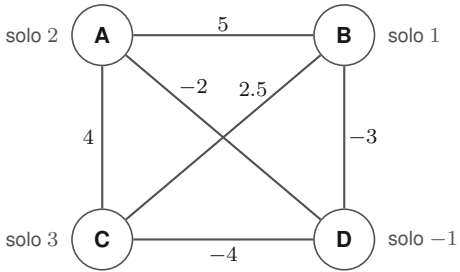
$$w(\{i,j\}) := u_{ij}^J = u_i(\{i,j\}) = u_j(\{i,j\}).$$

What it buys

- A likelihood; no selection rule
- Simulate who enters and who partners with whom

▸ Stable partition ▸ Proof sketch ▸ Uetake & Watanabe

Example: four firms, one stable partition



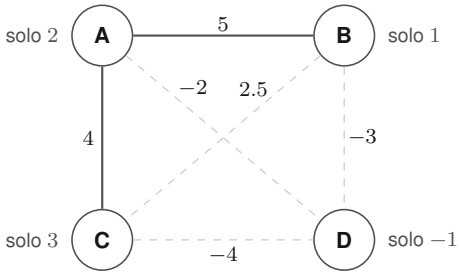
Next to a firm: payoff from bidding alone.

On a line: payoff to *each* member of that JV.

1. Four firms; every pair could form a JV

▸ Formal ranking definition

Example: four firms, one stable partition



Next to a firm: payoff from bidding alone.

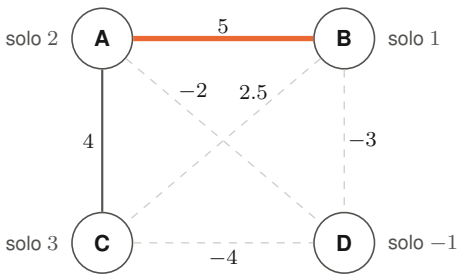
On a line: payoff to *each* member of that JV.

1. Four firms; every pair could form a JV
2. **Acceptable pairs**: only AB and AC

   Both members prefer the JV to bidding alone and to staying out.

▸ Formal ranking definition

Example: four firms, one stable partition



Next to a firm: payoff from bidding alone.

On a line: payoff to *each* member of that JV.

1. Four firms; every pair could form a JV
2. **Acceptable pairs**: only AB and AC
3. **Form the best one**: AB $(5 > 4)$

   One common w: set $w(ij) := u_{ij}^J$.

   All six pairs, ranked by w:

   $AB > AC > BC > AD > BD > CD.$

   Restrict this order to each firm's own pairs:

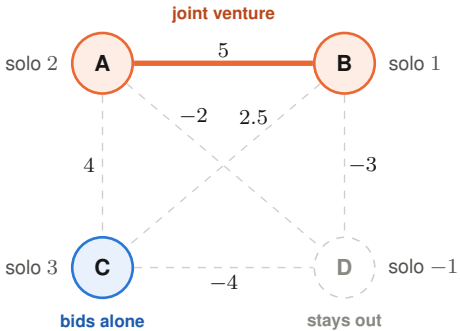
   $A \quad AB > AC > AD$
   $B \quad AB > BC > BD$
   $C \quad AC > BC > CD$
   $D \quad AD > BD > CD$

   Best: maximize w over acceptable pairs.

   $$\arg\max_{e \in \{AB, AC\}} w(e) = \{AB\} \qquad (5 > 4).$$

▸ Formal ranking definition

Example: four firms, one stable partition



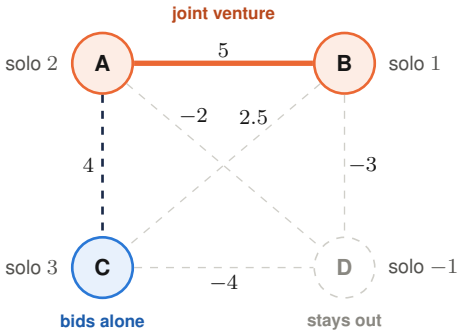
Next to a firm: payoff from bidding alone.

On a line: payoff to *each* member of that JV.

1. Four firms; every pair could form a JV
2. **Acceptable pairs**: only AB and AC
3. **Form the best one**: AB $(5 > 4)$
4. **Repeat**: C bids alone $(3 > 0)$, D stays out $(-1 < 0)$

▸ Formal ranking definition

Example: four firms, one stable partition



Next to a firm: payoff from bidding alone.

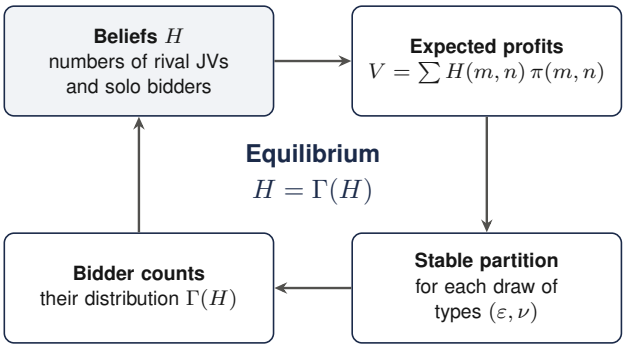
On a line: payoff to *each* member of that JV.

1. Four firms; every pair could form a JV
2. **Acceptable pairs**: only AB and AC
3. **Form the best one**: AB $(5 > 4)$
4. **Repeat**: C bids alone $(3 > 0)$, D stays out $(-1 < 0)$
5. **Stable**: AC does not block, A would lose $(4 < 5)$

Unique: any partition without AB is blocked by AB

▸ Formal ranking definition

Closing the model: beliefs about rivals must be consistent



- Stability takes beliefs as given
- **Equilibrium**: beliefs are correct
- Uniqueness of H is not proved

What the estimated model adds

	Ingredient	Motivated by
(a)	**Auction-level states**, seen by firms but not by us	JVs cluster in few auctions
(b)	**Direct effects of strength** on participation	entry varies with strength
(c)	**JV cost** by market and size; **availability** of pairs	partners repeat

31 parameters. All shocks are normal; solo and pair shocks are independent.

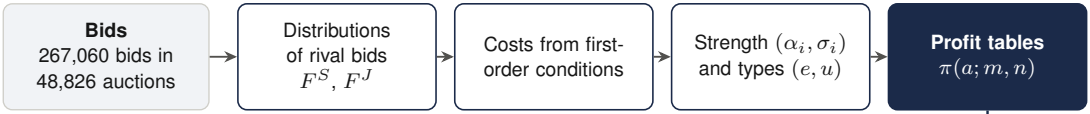
▸ Full payoffs

Roadmap

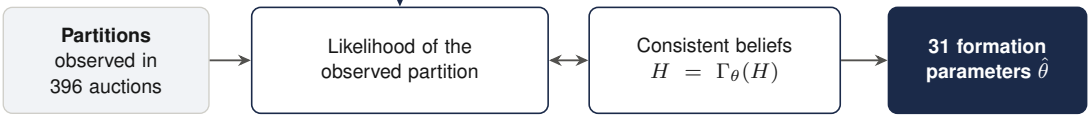
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Estimation in two steps

Step 1: Bidding



Step 2: Formation



- Step 1: bids only; a bid at the floor is a bound on the bidder's cost
- Step 2: who participates with whom, taking strength and the profit tables as given

▸ Step 1 in detail

Step 1: recovering costs and firm strength

For illustration: no price floor; all bids are interior.

1. Recover costs from bids: GPV (2000)

- Estimate the distributions of rival bids
- Invert the bidding first-order condition to recover costs $\widehat{c}_{it}$

2. Estimate firm and auction fixed effects: AKM (1999)

$$\log(\widehat{c}_{it}/R_t) = \delta_t + q_g(\log s_{it}) - \alpha_i + u_{it},$$
$$\log s_{it} = \mu_t + \sigma_i + e_{it}.$$

- Firm effects (α_i, σ_i): persistent cost and score strength
- Auction effects (δ_t, μ_t): common cost and score conditions

▸ When the floor binds

Step 2: maximum likelihood subject to consistent beliefs

$$\max_{\theta,\, H} \sum_t \log L_t(\theta, H) \qquad \text{subject to} \qquad H = \Gamma_\theta(H) \ \text{ in every auction}$$

$L_t(\theta, H)$: probability of observing partition P_t^{obs} in auction t.

The likelihood

- The observed partition must be stable

The constraint

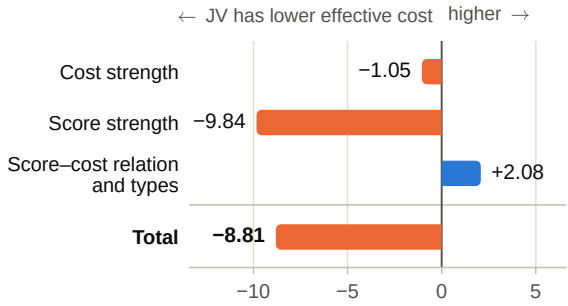
- Beliefs match the implied bidder counts

Algorithm: Nested Pseudo-Likelihood (NPL), Aguirregabiria and Mira (2007).

Conditional uniqueness makes both the likelihood and the counterfactuals computable.

JV advantage over a solo bidder with zero strength

A JV against a solo bidder with zero strength

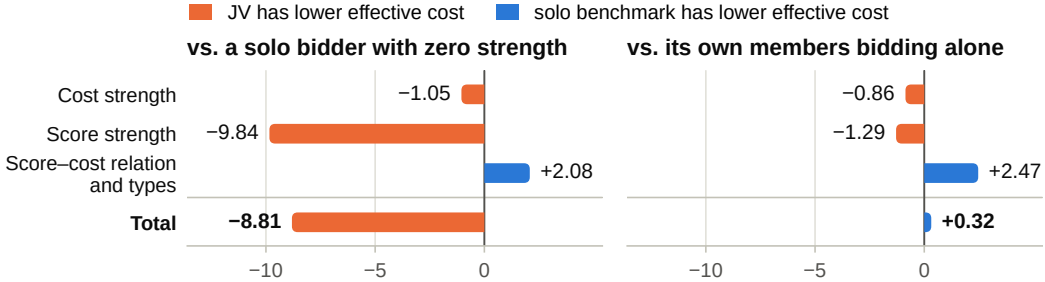


JV minus solo: mean log effective cost (log points)

- Effective cost: −**8.8 log points**
- Mainly from **higher scores**
- Little advantage from lower costs

▸ Against its own members ▸ Three benchmarks

No lower effective cost for the JV on average

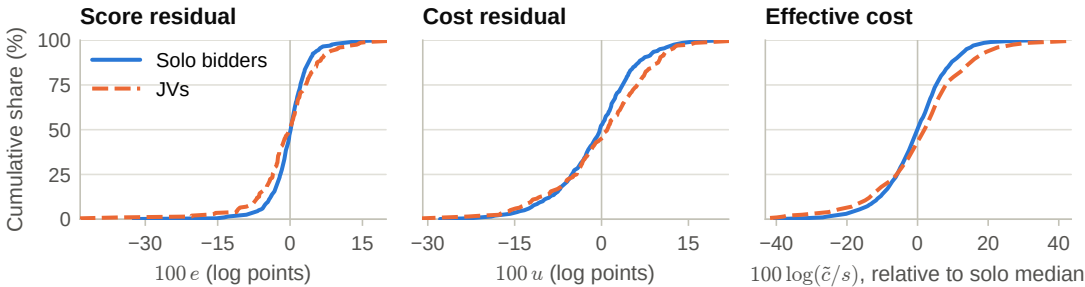


- **Right:** each JV versus the average of its two members evaluated as solo bidders
- Combining strengths: -2.15 log points; JV score–cost relation and types: $+2.47$
- **Net difference: $+0.32$ log points on average**

▸ Three benchmarks

JVs have more dispersed bidding outcomes

Effective cost: the same persistent strength in both modes ($\alpha = \sigma = 0$).



- JV score and cost residuals are **more dispersed**
- More very low effective-cost draws, and more very high ones

Capacity and project size shape participation

Effect on each firm's payoff	Alone	Jointly
Capacity shortfall	-1.81 (0.08)	-1.48 (0.33)
Project size	-0.67 (0.25)	$+0.40$ (0.14)

Units: solo-shock s.d. ($\hat{\sigma} = 197$ million yen). Clustered s.e. in parentheses.
JV shortfall uses pooled capacity; displayed effects are per member.

- **Capacity shortfalls discourage participation.**
  JVs pool the members' capacities, reducing the shortfall.
- **Beyond capacity, project size favors JVs.**
  The direct size term lowers solo payoffs and raises each member's JV payoff.

▸ All parameters ▸ Standard errors

Past joint bids lower the additional cost of a JV

Mean additional JV cost

Observed JVs	σ units	Billion yen
New pairs	6.31 (0.29)	$\mathbf{1.24}$ (0.31)
Repeat pairs	2.32 (0.26)	$\mathbf{0.46}$ (0.13)

Means over observed new and repeat JVs.

Totals for the pair: each member pays half.

Discount from previous joint bids

1 previous joint bid: **41% lower cost**

2 previous joint bids: **57% lower cost**

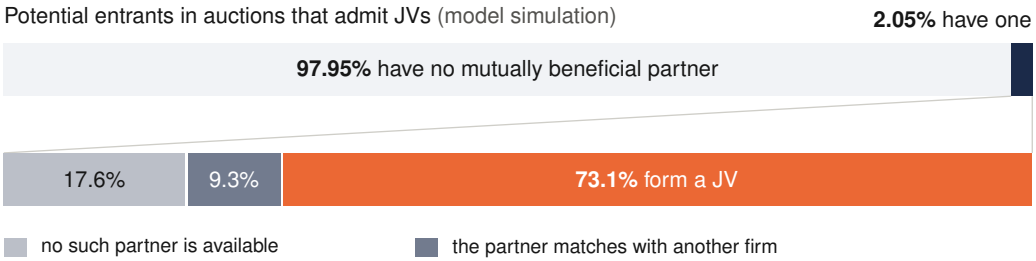
Pairs with no prior contact

19% probability of being available as partners.

No prior contact: never previously participated in the same auction.

▸ All parameters ▸ Standard errors

Mutually beneficial partners are scarce



- Mutually beneficial: *both* firms prefer the JV to bidding alone and to staying out
- JVs are limited mainly by the scarcity of attractive partnerships
- **Policy implication**: easier partner search alone may have limited impact.
  To promote JVs, lower **JV-specific entry costs** to make more partnerships attractive.

Fit

	Data	Model
Solo bids	6,159	6,694
JVs	312	353
new pairs	145	185
repeat pairs	167	168
Auctions with a JV	127	132
JVs per auction with a JV	2.46	2.66
Winning bid / reserve (%)	88.7	89.0
Winners within 1% of the floor (%)	74	77

- Solo bids are overstated by 9%
- The excess JVs are new partnerships
- **Weak spot**: six projects above 10 bn yen

	Data	Model
JVs	18	31.9
Repeat JVs	2	16.3
Winning bid / reserve (%)	94.1	90.2

The counterfactual therefore leaves out the six projects above 10 bn yen.

▸ By size and market ▸ Dispersion ▸ Prices

Roadmap

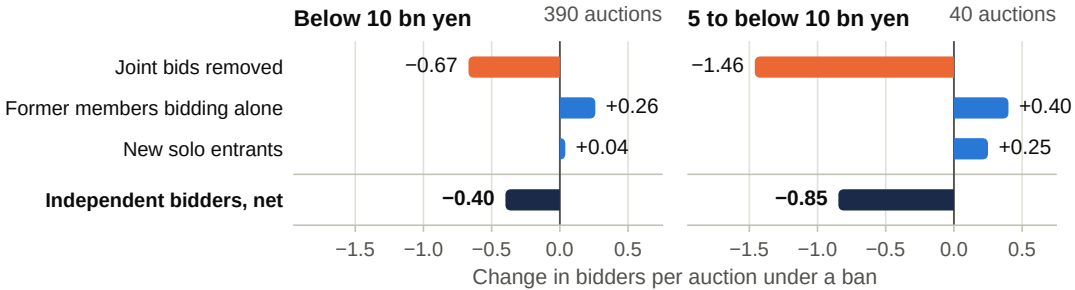
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Banning JVs: design

Policy	Ban JVs: **390 auctions below 10 bn yen**, of which **40 at 5 bn yen or more**
Responses	Beliefs solved again: former members may bid alone, other firms may enter
Outcomes	Winning **price** and **technical score**; probability of an award

▸ Outcome measures

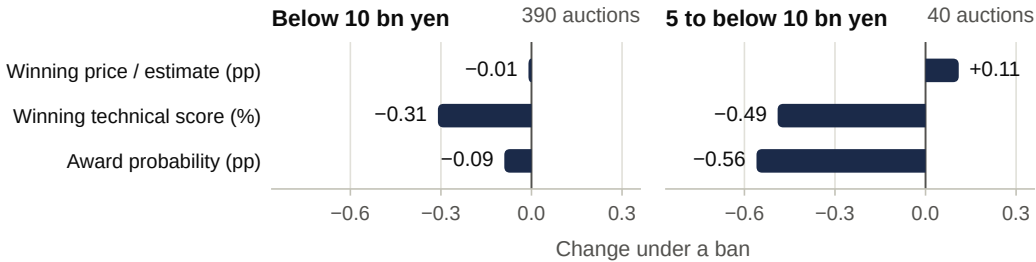
Under a ban, joint bids are only partly replaced



- **19%** of former JV members bid alone (14% in the larger projects); some outsiders enter
- Net loss of bidders: **2.4%** (7.0% in the larger projects)

▸ Full table ▸ Fixed profits and starting beliefs

A ban raises prices and lowers winning scores in the larger projects



- **5 to below 10 bn yen**: prices 89.55% → **89.66%** of the engineer's estimate; scores **−0.49%**
- **All 390 auctions**: prices 89.05% → 89.04%; scores −0.31%

▸ Full table ▸ Outcome measures

With a binding price floor, prices alone miss the effect of a ban

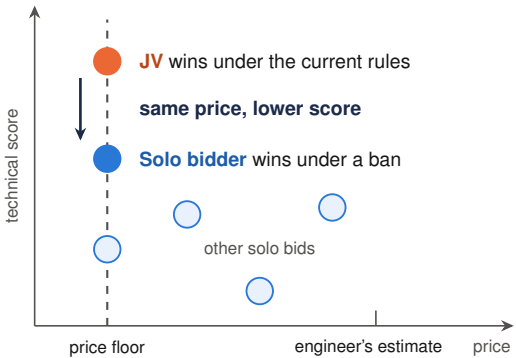


Illustration. The highest ratio of score to price wins.

- At the floor, the price cannot fall
- A higher score still wins more often
- A ban can lower scores at the same price

Discussion

- **Equal sharing** is an assumption: we do not observe the shares that partners agree on
- **Independent solo and pair shocks**: a strong assumption
- **Scope**: model predictions for the 390 auctions below 10 bn yen

Next: data on how partners share returns, and dependence between solo and joint participation.

▸ Fixed profits and starting beliefs ▸ Final optimization and standard errors

Conclusion

1. The effect of joint bidding depends on **members' outside options** and **rivals' responses**
2. **Equal sharing** makes formation tractable: a unique stable partition and a likelihood
3. In projects of 5 to below 10 bn yen, **allowing JVs lowers prices and raises technical scores**
4. With a binding price floor, **prices alone miss the effect**

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Backup slides

Related literature

This paper

Formation of mergers and JVs — pairs of firms; a unique stable outcome

Gowrisankaran (1999); Akkus et al. (2016); Fox (2018); Igami & Uetake (2020); Uetake & Watanabe (2020)

Joint bidding — who bids with whom; numbers vs. strength

Hendricks & Porter (1992); Iimi (2004); Estache & Iimi (2009); Branzoli & Decarolis (2015); Gugler et al. (2021)

Auctions with entry — entry partly offsets a ban

Li & Zheng (2009); Krasnokutskaya & Seim (2011); Athey, Levin & Seira (2011)

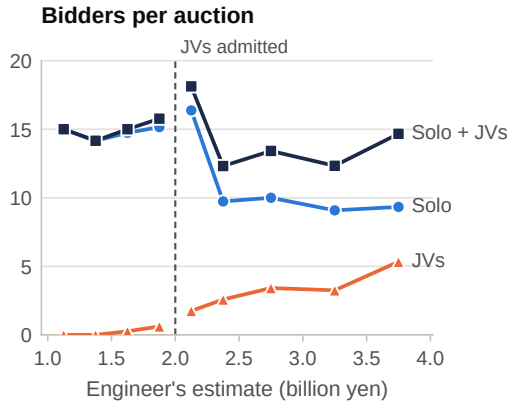
Scoring auctions and price floors — scores matter as well as prices

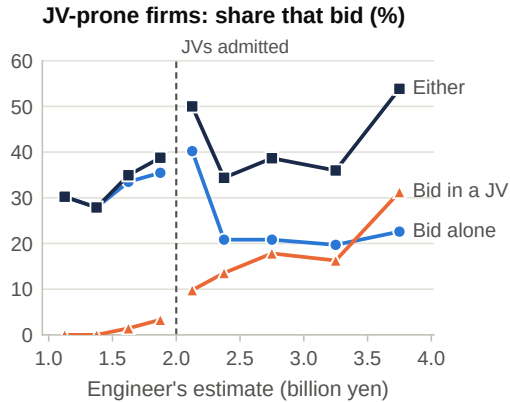
Che (1993); Asker & Cantillon (2008); Decarolis (2014)

Globally ranked pairs

Abraham et al. (2008)

Case study: Kanto bridges admit JVs above 2 billion yen





Kanto bridge auctions, 2006–2022; bin means. Right panel: the 47 firms that join a JV above the threshold.

At the threshold, firms switch from solo to joint bidding

	Below	Above	Jump (s.e.)
JV-prone firms (47), % of firm–auction pairs			
Bid alone	31.5	23.5	$-\mathbf{11.1}$ (4.2)
Bid in a JV	1.2	14.6	$+\mathbf{7.8}$ (3.0)
Participate	32.8	38.2	-3.3 (4.0)
Per auction (112 auctions, 1.25–3.5 bn yen)			
Solo bidders	14.59	10.69	-3.73 (1.62)
JVs	0.23	2.80	$+1.40$ (0.60)
Independent bidders	14.82	13.49	-2.33 (1.40)

Year effects and log size. S.e. clustered by auction.

- Changes **how** firms bid, not whether
- No change for the other 106 firms

1.42

displacement ratio $-\Delta$ solo / Δ JV

95% confidence set $[0.48, \infty)$

◂ Back ▸ Specification ▸ Robustness

Bidding: from a bidder's problem to expected profits

Facing m JVs and n solo rivals, a bidder with score s and cost $\tilde{c} = c/R$ chooses $y = \log\left(s/\frac{b}{R}\right)$:

$$\max_{\log s \le y \le \log(s/\tau)} \underbrace{R\left(s\,e^{-y} - \tilde{c}\right)}_{\text{profit if it wins}} \underbrace{F^J(y)^m\, F^S(y)^n}_{\text{probability of winning}}$$

- Only the **effective cost** $\tilde{c}/s$, cost per unit of score, matters for the bid
- At the floor, a range of costs leads to the same bid
- Bidders know how many rivals of each kind they face, not who they are

Output: expected profits $\pi_i^S(m,n)$ and $\pi_{ij}^J(m,n)$ for every firm and every pair.

A stable partition: no firm and no pair wants to deviate

A **partition** assigns every firm to one option. U_i: firm i's payoff in it.

Definition. A partition is **stable** if

(i) no firm prefers to stay out or to bid alone: $U_i \geq \max\{0,\, u_i^S\}$ for every i;

(ii) no *blocking pair*: no available pair (i, j) with $u_{ij}^J > U_i$ and $u_{ij}^J > U_j$.

- A firm can leave its partner on its own; a new JV needs both members to agree
- Core stability, restricted to single firms and pairs (Ray & Vohra 2015)
- Firms hold the beliefs H fixed when they consider a deviation

What the full model adds

$$u_{it}^{S} = \frac{V_{it}^{S}(\xi_t, \eta_t; H)}{\sigma} - f_t - \lambda_S\, g_{it} - \beta_S\, \ell_t \underbrace{+ \kappa_{S,\alpha}\, \tilde{\alpha}_i + \kappa_{S,\sigma}\, \tilde{\sigma}_i}_{\text{(b)}} \underbrace{+ c_S\, \xi_t}_{\text{(a)}} + \varepsilon_{it}$$

$$u_{ijt}^{J} = \frac{1}{2}\Big[\frac{V_{ijt}^{J}(\xi_t, \eta_t; H)}{\sigma} - K_t\, e^{-\gamma h_{ijt}} - \lambda_J\, g_{ijt}\Big] - f_t - \beta_J\, \ell_t \underbrace{+ \kappa_{J,\alpha}\, \tilde{\alpha}_{ij} + \kappa_{J,\sigma}\, \tilde{\sigma}_{ij}}_{\text{(b)}} \underbrace{+ c_J\, \eta_t}_{\text{(a)}} + q\, \nu_{ijt}$$

(a) **Auction-level states** (ξ_t, η_t): seen by firms, not by us

(b) **Direct effects of strength** $\tilde{\alpha}$, $\tilde{\sigma}$, beyond expected profits

(c) **Availability**: a pair without prior contact is available with probability $e^{-\tau}$

$g = \max\{0, \log(\text{size}/\text{capacity})\}$; $\ell_t = \max\{0, \log(R_t/3 \text{ bn yen})\}$; f_t, K_t by market and size; q fixed at 1.

Step 1: costs and strength when the floor binds

1. **Rival bids**: normal with auction effects
2. **Costs**: invert the first-order condition
3. **Strength**: within-auction comparisons
4. **Types** and **profit tables**

What a bid reveals about cost



27% at the floor

14% at or above the reserve

of the solo bids we compare

Censored bids enter as bounds on cost: dropping them would bias strength.

Step 2: the likelihood of the observed partition

By Proposition 1, the likelihood is the probability that the observed partition is **stable**.

Given the auction's states (ξ, η) and the payoffs U of the observed bidders:

Stability requires that	**Factor in the likelihood**
non-participants prefer to stay out	$\Phi(-a_i)$
solo bidders prefer to bid alone	$\Phi(a_i)$
each JV is available and beats its members' own options	$p_{ij}\,\Phi(\bar{m}_{ij}/s)\,\Phi(U_{ij}-a_i)\,\Phi(U_{ij}-a_j)$
no other available pair blocks	$1-p_{ij}+p_{ij}\,\Phi\big((\max\{U_i,U_j\}-\bar{m}_{ij})/s\big)$

$a_i, \bar{m}_{ij}$: mean payoffs of bidding alone and of the pair; p_{ij}: probability that the pair is available; s: s.d. of the pair shock.

$$L_t(\theta, H) = \iint \Pr\big(P_t^{\text{obs}} \text{ is stable} \mid \xi, \eta; \theta, H\big)\, \phi(\xi)\, \phi(\eta)\, d\xi\, d\eta$$

Only the observed bidders' payoffs are simulated; millions of other pairs enter in closed form.

◂ Back ▸ Exact expression

Backup

What variation pins down the parameters

Parameter	Variation that moves it
Participation cost f	participation of firms with no capacity shortfall
Capacity λ_S, λ_J	how participation falls with the shortfall
Strength κ	participation by strength, at given expected profits
History discount γ	how much more often past partners form JVs
Availability $e^{-\tau}$	JVs without prior contact: 16 of 312 (80% of pairs lack contact)
State scales c_S, c_J	dispersion of solo bidders; clustering of JVs
Yen scale σ	response of participation to expected profits (weak)
JV cost K, shock scale q	JVs are rare: 312 among 6.68 million pairs

K and q are hard to separate: we fix $q = 1$, and cost levels are conditional on it.

Backup: size thresholds for project-specific JVs

Facility	JVs allowed above	With special technology
Bridges	2 bn yen	1 bn yen
Buildings	3 bn yen	1.5 bn yen
Weirs, sluices, pumping stations, tunnels, utility tunnels	5 bn yen	2.5 bn yen
Dams	10 bn yen	5 bn yen

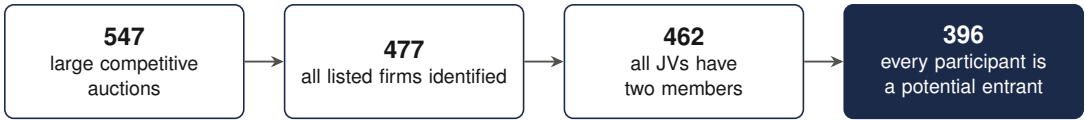
- MLIT circular on JVs (1988, last revised 2006): guidelines; each bureau decides
- Two or three members; in a two-firm JV each member holds at least 30%
- A JV is examined as a single bidder

Backup: estimation sample

	Auctions	Median size (bn yen)	Potential entrants	Solo bidders	JVs	With a JV (%)	JVs: new / repeat
Kanto, civil engineering	69	3.53	117	14.1	0.96	43	48 / 18
Kanto, bridges	58	2.24	49	13.4	2.31	69	34 / 100
Kinki, civil engineering	99	3.06	130	18.3	0.24	12	18 / 6
Kyushu, civil engineering	50	2.89	63	17.8	0.74	40	22 / 15
Other bridges	104	0.66	38	14.7	0.29	18	16 / 14
Other civil engineering	16	4.77	41	11.6	1.31	38	7 / 14
Size quartile 1 (0.38–1.53)	99	0.62	39	14.8	0.21	14	9 / 12
Size quartile 2 (1.58–2.44)	99	2.00	313	16.6	0.68	29	31 / 36
Size quartile 3 (2.45–3.63)	99	2.92	129	16.9	0.89	38	30 / 58
Size quartile 4 (3.63–30.56)	99	4.95	59	13.9	1.37	46	75 / 61
$\geq$ 5 bn yen	46	5.75	56	9.7	1.72	63	54 / 25
$\geq$ 10 bn yen	6	20.08	46	0.7	3.00	100	16 / 2
All	396	2.45	58	15.6	0.79	32	145 / 167

Potential entrants: median. Solo bidders and JVs: means per auction. Repeat JV: its members bid together before.

Backup: sample construction and JV members



- **JV members**: names linked to firms; all members linked for 499 of 533 JV bids (93.6%)
- **Potential entrants**: three-year window and one-tenth size screen
- Two-year window: 384 auctions. One-quarter size screen: 364 auctions
- Looser screens: 398–400 auctions, but median sets of 85–445 firms instead of 57

Backup: collusion screens

- Collusion is a known concern (Kawai & Nakabayashi 2022; Chassang et al. 2022)
- **Marginal-bid screen**: no discontinuity as effective bids converge, by bureau and by year
- **Five standard screens**, 56 bureau-by-category cells: only Shikoku bridges is flagged
- It contributes 10 of the 396 auctions; we keep it, and treat bidding as competitive

Backup: threshold regressions

Kanto bridge auctions, 2006–2022; cutoff $c = 2$ bn yen; main window 1.25–3.5 bn yen.

$$\text{Auction level:} \quad Y_t = a + \beta D_t + \theta \log(R_t/c) + \lambda_{y(t)} + u_t, \qquad D_t = \mathbf{1}\{R_t \geq c\}$$

$$\text{Firm level:} \quad P_{it}^m = \alpha_i^m + \beta^m D_t + \theta^m \log(R_t/c) + \lambda_{y(t)}^m + \varepsilon_{it}^m, \qquad m \in \{\text{solo}, \text{JV}, \text{any}\}$$

- Year effects, a common slope in log size, firm fixed effects; s.e. clustered by auction
- Potential entrants: firms that bid on a Kanto bridge within two years
- JV-prone firms: the 47 that join a JV above the threshold; 106 other firms
- $\hat{d} = -\hat{\beta}^{\text{solo}}/\hat{\beta}^{\text{JV}}$; confidence set from the test of $\beta^{\text{solo}} + d\,\beta^{\text{JV}} = 0$ (Fieller)

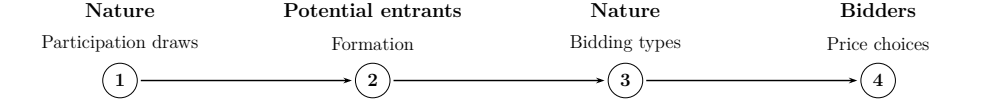
Backup: the threshold estimates are robust in windows near the cutoff

		Per auction		JV-prone firms (probability)		
Sample	Auctions	Solo bidders	JVs	Bid alone	Bid in a JV	$\hat{d}$
Main: 1.25–3.5 bn yen	112	-3.73 (1.62)	$+1.40$ (0.60)	-0.111 (0.042)	$+0.078$ (0.030)	1.42
1.5–3.0 bn yen	75	-2.83 (1.98)	$+0.52$ (0.68)	-0.052 (0.044)	$+0.032$ (0.034)	1.62
1.0–4.0 bn yen	158	-4.50 (1.61)	$+1.75$ (0.46)	-0.114 (0.040)	$+0.089$ (0.024)	1.28
Cutoff at 2.1 bn yen	112	-3.53 (1.60)	$+1.24$ (0.64)	-0.118 (0.044)	$+0.067$ (0.033)	1.75
Excluding 1.9–2.2 bn yen	101	-5.19 (2.14)	$+2.07$ (0.83)	-0.156 (0.056)	$+0.116$ (0.042)	1.35
Steel bridges only	101	-3.24 (1.62)	$+1.79$ (0.66)	-0.090 (0.037)	$+0.105$ (0.033)	0.85
2006–2013	79	-3.40 (1.85)	$+1.68$ (0.67)	-0.116 (0.045)	$+0.081$ (0.032)	1.43
2014–2022	33	-3.66 (3.41)	$+0.60$ (1.26)	-0.074 (0.110)	$+0.055$ (0.069)	1.34
Formation sample, 1.58–5.1 bn	58	-1.77 (1.53)	$+1.28$ (0.70)	-0.041 (0.037)	$+0.067$ (0.034)	0.61

Jumps at the threshold (s.e. clustered by auction). $\hat{d}$: minus the solo jump over the JV jump.

Backup: timing and information in the full model

Given auction t: R_t (reserve), $\tau_t R_t$ (price floor), $\mathcal{N}_t$ (potential entrants).
Known before formation: $I_t = (X_t, \mu_t, \delta_t)$ (common auction conditions) and own $a_i = (\alpha_i, \sigma_i)$ (bidding strength).



Draw (ξ_t, η_t):
common solo/JV payoff states.

Draw $\varepsilon_{it}, \nu_{ijt}$:
solo and pair payoff shocks.

Partnership opportunities
$\longrightarrow \mathcal{A}_{it}$: available partners.

Observe (ξ_t, η_t), $\mathcal{A}_{it}$, and all option payoffs u^S_{it}, u^J_{ijt}.

Stay out, bid alone, or agree to form a JV.
$\longrightarrow$ Stable partition P; M_t JVs, N_t solo bidders.

Future types and rival counts are unknown when choosing.

Draw solo types
$(e_{it}, u_{it}) \sim G^S_g(\cdot \mid I_t)$

and JV types
$(e_{ijt}, u_{ijt}) \sim G^J_g(\cdot \mid I_t)$.

$\longrightarrow$ Each bidder learns its own score s_i and cost c_i.

Observe bidder counts (M_t, N_t); rivals' strengths and types remain unknown.

Submit $b_i \in [\tau_t R_t, R_t]$.

Buyer awards the contract to the highest s_i/b_i.
The winner earns $b_i - c_i$.

Backup: first-order condition and boundary bids

An interior solution satisfies

$$\frac{1}{1-\tilde{c}\,e^{y}/s} \;=\; n\,\frac{f^{S}(y)}{F^{S}(y)} + m\,\frac{f^{J}(y)}{F^{J}(y)} \;\equiv\; h(y),$$

which depends on the bidder's type and strength only through the effective cost $\tilde{c}/s$.

- **Floor.** A bid below $(1+\epsilon)\tau R$, $\epsilon = 0.01$, reveals only $\tilde{c} \le (1+\epsilon)\,\tau\left(1 - 1/h(y^{\epsilon})\right)$
- **Reserve.** A bid at or above the reserve reveals only $\tilde{c} \ge 1 - 1/h(y^{R}), \quad y^{R} = \log s$
- A lone bidder bids the reserve
- Simplification: bidders observe the numbers of rivals when they bid

Backup: proof sketch of Proposition 1

$\mathcal{E}^*$: the available pairs that both members prefer to their own best individual option.

1. In a stable partition every JV belongs to $\mathcal{E}^*$ (condition (i))
2. **The best pair is unavoidable**: apart, each member gets strictly less, so the pair blocks
3. **Induction**: remove the pair; the argument applies to the remaining firms
4. **Existence**: the greedy outcome is stable; a blocking pair would have been chosen earlier

Uses the common pair ranking and no payoff ties.
Unequal sharing need not preserve this ranking.

Backup: relation to Uetake & Watanabe (2020)

Both papers model entry and partner choice jointly, and require mutual consent.

	Uetake & Watanabe (2020)	This paper
Who matches	entrants with incumbents	any two potential entrants
Payoffs	two sides value a merger differently	the same payoff for both members
Stable outcomes	multiple; extremal matchings as bounds	unique given beliefs; a likelihood
Competition	a conjectured number of operating firms	beliefs over rival numbers, $H = \Gamma(H)$

Backup: pair shocks and the number of potential partners

$$\nu_{ijt} = r_t\,\zeta_{ijt} - l_t, \qquad \zeta_{ijt} \sim N(0,1)$$

- **Problem**: more partners, more exceptional draws: $\Pr(\max_{j \le m} \zeta_j \le z) = \Phi(z)^m$
- At least one draw above 2: 21% with 10 partners, 90% with 100
- **Adjustment**: r_t, l_t equalize the location and spread of the best pair shock (cf. Menzel 2015)
- Fixed before estimation: $r_t \in [0.83, 1.16]$, $l_t \in [-0.73, 0.84]$
- **Scale**: q is the s.d. of the pair shock relative to the solo shock. Main specification: $q = 1$

Backup: distributions of rival bids

$$F^d(y \mid X_t, \mu_t, \delta_t, m, n) = \Phi\left(\frac{y - \mu^d(X_t, m, n) - \omega_t}{\sigma_t^d}\right), \qquad \omega_t \sim N(0, \sigma^2_{\omega,t}), \qquad d \in \{S, J\}$$

- ω_t: a shift common to solo and JV bids in the auction (Krasnokutskaya 2011)
- Mean: market effects, year trend, log floor ratio, quadratic in $(\log R_t, m, n)$
- Maximum likelihood on 267,060 bids (261,400 solo, 5,660 JV) in 48,826 auctions
- Within-auction s.d.: 0.057 (solo), 0.059 (JV) at the median; s.d. of the shift: 0.046 in 2014
- An approximation to an equilibrium object; it does not impose normal types

Backup: estimating strength from censored costs

Two solo bidders i and j in one auction: $\Delta = q(\log s_{it}) - q(\log s_{jt}) - (\alpha_i - \alpha_j)$.

- Trim both log costs to the range where *both* are observed: $E\big[m_{ijt}(\Delta_0)\,(w_{it} - w_{jt})\big] = 0$
- The auction's cost condition δ_t differences out; no distribution is needed for u
- Honoré (1992); two-sided censoring as in Alan, Honoré, Hu & Leth-Petersen (2014)
- Empirical Bayes shrinkage; 1,086,696 pairs of 7,728 firms in 40,150 auctions
- **Scores** are not censored: within-auction least squares gives σ_i
- **JVs**: the score loads 0.86 (0.33) on the higher member strength, -0.28 (0.25) on the lower

Backup: Monte Carlo for the strength estimator

	Score coefficient	$\hat{\sigma}_\alpha$	Slope, true on estimated			Correlation		
			1–4 bids	5–19	≥ 20	1–4 bids	5–19	≥ 20
Estimator	0.49	0.035	0.95	0.96	0.97	0.67	0.87	0.95
Known rival distributions	0.48	0.034	1.01	1.01	1.00	0.70	0.89	0.96
Reserve bids as interior	0.36	0.029	1.18	1.19	1.19	0.66	0.86	0.95
Floor bids dropped	0.16	0.025	1.34	1.34	1.37	0.53	0.77	0.92
Floor bounds as costs	0.05	0.021	1.63	1.61	1.63	0.58	0.81	0.93
Truth	0.50	0.035		1				

- 12,000 simulated auctions, 4,000 firms; 28% of bids at the floor and 11% at the reserve
- The estimator recovers the score coefficient and the dispersion; ignoring censoring does not

Backup: the conditional probability of a partition

Write $u^S_{it} = a_{it} + \varepsilon_{it}$ and $u^J_{ijt} = \bar{m}_{ijt} + s_t\,\zeta_{ijt}$, with $s_t = q\,r_t$. For $P^{\text{obs}}_t = (S_t, J_t, O_t)$:

$$C_t = \prod_{i \in O_t} \Phi(-a_{it}) \prod_{i \in S_t} \Phi(a_{it}) \prod_{(i,j) \in J_t} p_{ij}\,\Phi\Big(\frac{\bar{m}_{ijt}}{s_t}\Big)$$

$$p_t\big(P^{\text{obs}}_t \mid \xi, \eta; \theta, H\big) = C_t\,\mathbb{E}_+\Bigg[\prod_{(i,j) \in J_t} \Phi(U_{ij} - a_{it})\,\Phi(U_{ij} - a_{jt}) \prod_{(i,j) \in \mathcal{E}_t \setminus J_t} B_{ij}\big(\max\{U_i, U_j\}\big)\Bigg]$$

$$B_{ij}(v) = 1 - p_{ij} + p_{ij}\,\Phi\Big(\frac{v - \bar{m}_{ijt}}{s_t}\Big)$$

- $\mathbb{E}_+$: over the observed bidders' payoffs, each conditional on being positive
- One draw per solo bidder, one shared draw per JV; non-participants have $U_i = 0$
- p_{ij}: 1 with prior contact, $e^{-\tau}$ without, 0 if JVs are not admitted
- 128 pairs of states per auction; scrambled Sobol points for the payoffs

Backup: the parameter and belief iteration

1. Beliefs at 25 points of (ξ, η)
2. Profits from current beliefs
3. Parameter step
4. Two belief updates, damping 0.5
5. Five rounds of 2–4
6. 16 updates, damping 0.1; a final step

Not a proof of convergence.

Round	Log lik.	σ (m yen)	$\kappa_{S,\sigma}$	$\kappa_{J,\sigma}$	c_S	c_J
1	$-10{,}659.6$	174	0.59	0.39	0.64	0.44
2	$-10{,}660.7$	218	0.60	0.40	0.63	0.46
3	$-10{,}663.3$	237	0.60	0.40	0.63	0.46
4	$-10{,}660.6$	239	0.60	0.40	0.63	0.45
5	$-10{,}665.5$	256	0.60	0.41	0.63	0.45
Final	$-10{,}656.1$	204	0.59	0.39	0.64	0.43

Remaining gaps (mean absolute, total rivals)

	Solo	JV
All grid points (maximum)	0.36	0.58
Six largest projects	1.07	1.84

Final row: the parameters used in all simulations.

Backup: the final optimization and standard errors

- **Final optimization**: the 31 coefficients once more, at fixed profit and belief tables
- Log likelihood $-10{,}656.14 \to -10{,}651.18$; β_S: $0.97 \to 0.67$; $\bar{f}$: $64.27 \to 2.20$
- **Standard errors**: auction-clustered sandwich, $T = 396$ auctions, analytic scores $\hat{g}_t$

$$\widehat{V} = \widehat{A}^{-1}\Big\{\frac{T}{T-1}\sum_{t=1}^{T}(\hat{g}_t - \bar{g})(\hat{g}_t - \bar{g})'\Big\}\widehat{A}^{-1}, \qquad \widehat{A} = -\sum_{t}\nabla_\theta\,\hat{g}_t$$

- **Conditional** on $q = 1$, the bidding estimates and the belief tables
- **Simulations** (fit, partner classification, the ban) use the parameter vector *before* this step

Backup: all formation parameters

Bidding alone	Estimate	(s.e.)
Participation cost, base cell $\bar{f}$	2.200	(0.817)
log shift: Kanto bridges	−1.058	(0.140)
log shift: Kinki civil eng.	−0.198	(0.088)
log shift: Kyushu civil eng.	−0.438	(0.171)
log shift: other bridges	−1.285	(0.353)
log shift: other civil eng.	−0.621	(0.379)
log shift: size quartile 2	−0.338	(0.360)
log shift: size quartile 3	−0.512	(0.366)
log shift: size quartile 4	−0.757	(0.402)
Capacity shortfall, λ_S	1.808	(0.077)
Project size, β_S	0.669	(0.246)
Cost strength, $\kappa_{S,\alpha}$	−0.176	(0.028)
Score strength, $\kappa_{S,\sigma}$	0.603	(0.031)
Solo state s.d., c_S	0.636	(0.028)
Expected profit, $1/\sigma$	0.00508	(0.00122)

Bidding jointly	Estimate	(s.e.)
JV cost K, size quartile 1	7.198	(0.920)
JV cost K, size quartile 2	5.941	(0.440)
JV cost K, size quartile 3	6.095	(0.343)
JV cost K, size quartile 4	5.901	(0.478)
log shift: Kanto bridges	0.015	(0.061)
log shift: Kinki civil eng.	0.208	(0.054)
log shift: Kyushu civil eng.	0.115	(0.067)
log shift: other bridges	−0.085	(0.112)
log shift: other civil eng.	0.110	(0.097)
Discount for past joint bids, γ	0.771	(0.073)
Contact parameter, τ	1.643	(0.400)
Capacity shortfall, λ_J	2.959	(0.657)
Project size, β_J	−0.401	(0.136)
Cost strength, $\kappa_{J,\alpha}$	−0.323	(0.062)
Score strength, $\kappa_{J,\sigma}$	0.397	(0.068)
JV state s.d., c_J	0.424	(0.052)

31 parameters; $\log q = 0$ fixed. $1/\sigma$ per million yen. Reference cell: Kanto civil engineering, size quartile 1 (it has no auctions). Log likelihood $-10{,}651.2$; 396 auctions. Auction-clustered s.e., conditional on $q = 1$, the bidding estimates and the belief tables.

Backup: effective-cost decomposition with three solo benchmarks

Contribution (log points)	Solo benchmark: Zero strength	Solo benchmark: JV members individually	Solo benchmark: Observed solo bidders
Cost strength α	-1.05	-0.86	-0.94
Score strength σ	-9.84	-1.29	-0.95
Score–cost relation and types (q, G)	$+2.08$	$+2.47$	$+2.56$
Total difference, JV minus solo	$\mathbf{-8.81}$	$\mathbf{+0.32}$	$\mathbf{+0.66}$
Auctions	127	127	122
Observed JVs	312	312	302

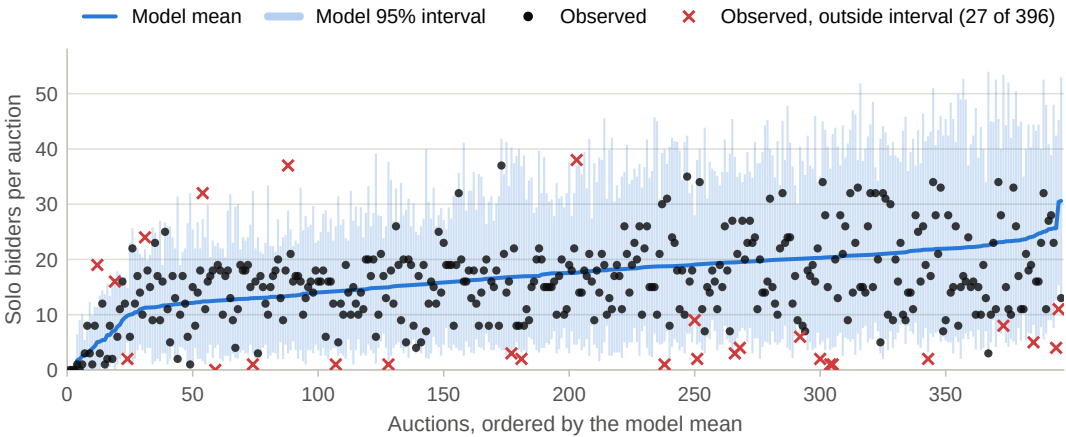
- Differences in $100\, E[\log(\tilde{c}/s)]$; negative: a lower effective cost for the JV
- Contributions are averaged over the two orders of the decomposition
- Estimates and types are held fixed; no confidence intervals

Backup: fit by project size and market

		Solo bids		JVs		New / repeat JVs		With a JV	
	Auctions	Data	Model	Data	Model	Data	Model	Data	Model
All	396	6,159	6,694	312	353	145 / 167	185 / 168	127	132
Size quartile 1	99	1,463	1,561	21	25	9 / 12	13 / 12	14	15
Size quartile 2	99	1,643	1,795	67	72	31 / 36	34 / 38	29	28
Size quartile 3	99	1,676	1,893	88	71	30 / 58	33 / 38	38	32
Size quartile 4	99	1,377	1,445	136	184	75 / 61	104 / 80	46	57
$\geq$ 5 bn yen	46	447	490	79	107	54 / 25	60 / 47	29	29
$\geq$ 10 bn yen	6	4	7	18	32	16 / 2	16 / 16	6	6
Kanto, civil engineering	69	971	1,052	66	120	48 / 18	70 / 50	30	36
Kanto, bridges	58	779	914	134	106	34 / 100	34 / 72	40	32
Kinki, civil engineering	99	1,807	1,904	24	30	18 / 6	16 / 14	12	19
Kyushu, civil engineering	50	889	938	37	45	22 / 15	34 / 11	20	19
Other bridges	104	1,527	1,617	30	35	16 / 14	22 / 13	19	19
Other civil engineering	16	186	268	21	17	7 / 14	9 / 8	6	7

Model: means over 64 simulated stable partitions per auction, with the auction-level states drawn from their distribution.

Backup: the model captures the dispersion of solo participation



- 27 of the 396 auctions (6.8%) lie outside the model's central 95% predictive interval
- S.d. of observed minus predicted solo bidders: 7.6; root mean predictive variance: 8.4

Backup: prices and the floor, model and data

		Winning bid / engineer's estimate (%)		Winners within 1% of the floor (%)	
	Auctions	Data	Model	Data	Model
All	396	88.7	89.0	74	77
$\geq$ 5 bn yen	46	90.3	89.7	70	81
$\geq$ 10 bn yen	6	94.1	90.2	33	50
Size quartile 1	99	88.5	89.1	72	70
Size quartile 2	99	88.6	88.9	71	72
Size quartile 3	99	88.4	88.6	79	83
Size quartile 4	99	89.5	89.3	76	83

- States at their posterior means; correlation of simulated and observed price ratios: 0.74
- In the largest projects the model predicts lower prices and a floor that binds more often

Backup: banning JVs, all measures

	All below 10 bn yen (390 auctions)	5 to below 10 bn yen (40 auctions)
Current policy		
Solo bidders per auction	15.75	10.69
JVs per auction	0.67	1.46
Contracts won by a JV (%)	7	18
Winning price / engineer's estimate (%)	89.05	89.55
Award probability (%)	99.75	99.47
Under the ban		
Members bidding alone (%)	19	14
New solo entrants per auction	0.04	0.25
Δ independent bidders per auction	-0.399	-0.847
Winning price / engineer's estimate (%)	89.04	89.66
Δ winning price / engineer's estimate (pp)	-0.01	$+0.11$
Winning score change (%)	-0.31	-0.49
Δ award probability (pp)	-0.09	-0.56

64 formation draws and 16 bidding replications per auction, common to both policies. The 40 auctions are among the 390.

◂ Back to who bids ◂ Back to prices and scores

Backup: how the outcomes of the ban are measured

$k = 0$: current rules; $k = 1$: ban. $\mathbb{E}_t$: mean over simulation draws. $\mathcal{T}$: a group of auctions.
S_t^k, J_t^k, O_t^k: solo bidders, JVs and non-participants; $N_t^k = |S_t^k|$, $M_t^k = |J_t^k|$; L_t^0: current JV members.

$$\text{Members bidding alone} = 100\,\frac{\sum_{t\in\mathcal{T}} \mathbb{E}_t\,|L_t^0 \cap S_t^1|}{\sum_{t\in\mathcal{T}} \mathbb{E}_t\,|L_t^0|} \qquad \text{New solo entrants} = \frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} \mathbb{E}_t\,|O_t^0 \cap S_t^1|$$

$$\Delta\,\text{Independent bidders} = \frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} \mathbb{E}_t\big[N_t^1 - N_t^0 - M_t^0\big] \qquad \text{Award probability}^k = \frac{100}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} \mathbb{E}_t\big[A_t^k\big]$$

$$\text{Price}^k = 100\,\frac{\sum_{t\in\mathcal{T}} \mathbb{E}_t\big[B_t^k\big]}{\sum_{t\in\mathcal{T}} R_t\,\mathbb{E}_t\big[A_t^k\big]} \qquad \Delta\,\text{Score} = \frac{100}{|\mathcal{T}|}\sum_{t\in\mathcal{T}}\Big(\frac{\bar{s}_t^1}{\bar{s}_t^0} - 1\Big)$$

- $A_t^k = 1$ if a usable bid is submitted; B_t^k: the winner's price; $\bar{s}_t^k$: mean winning score given an award
- Prices are weighted by the engineer's estimate; score changes are within-auction percentages

Back to prices and scores

Backup: fixed expected profits and alternative starting beliefs

	All below 10 bn yen (390 auctions)	5 to below 10 bn yen (40 auctions)
Δ *independent bidders per auction*		
Expected profits held at their current-policy values	-0.415	-1.118
Beliefs updated under the ban (main)	-0.399	-0.847
Δ *winning price / engineer's estimate (pp)*		
Starting from the estimation beliefs (main)	-0.013	$+0.112$
Starting from beliefs of comparable auctions	-0.006	$+0.141$

- Equilibrium adjustment matters more in the larger projects
- Other starting beliefs: the signs are unchanged; the largest difference is 0.029 points
- A check on initialization, not on uniqueness of the equilibrium

Back to who bids